CHAPTER 6

CORRELATION

CHAPTER ESSENCE AT A GLANCE:
Quantify bivariate relationships: positive versus negative correlation, scatter diagrams, Karl Pearson coefficient of correlation (r), and Spearman rank correlation coefficient (r_s).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Bivariate Relationship Dynamics:** Correlation studies and measures the direction and strength of association between two quantitative variables without proving direct cause and effect.

• **Scatter Diagram Visual Topology:** Graphic representation where paired points (X, Y) are plotted; upward sloping straight band indicates perfect positive correlation ($r = +1$); circular cloud indicates zero correlation ($r = 0$).

• **Karl Pearson Product Moment Coefficient (r):** Mathematical measure bounded strictly between -1 and +1 (-1 to +1), independent of change of origin and change of scale.

• **Spearman Rank Correlation (r_s):** Non-parametric correlation based on ordinal ranks assigned to qualitative attributes (beauty, judgment) using difference of ranks $d = R_1 - R_2$.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2\sum(Y-\bar{Y})^2}} \quad [-1 \leq r \leq +1]$$

Karl Pearson Product Moment Correlation Coefficient

$$r_s = 1 - \frac{6\sum D^2}{N(N^2-1)} \quad [D = R_X - R_Y]$$

Spearman Rank Correlation Coefficient (Untied Ranks)

$$\text{Correlation Properties} \Longleftrightarrow r_{u,v} = r_{x,y} \quad [\text{Invariant to Shift of Origin ¤ext\& Scale}]$$

Linear Invariance of Pearson Correlation

$$\text{Correlation Limits} \Longleftrightarrow -1\ (\text{Perfect Negative}) \leq r \leq +1\ (\text{Perfect Positive})$$

Bivariate Mathematical Bounding Limits

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Correlation does NOT imply causation; two variables may show high correlation due to a third mutual factor or sheer statistical coincidence (nonsense correlation).

• **Alert:** The value of Karl Pearson's correlation coefficient can NEVER exceed $+1$ or be less than -1; if your calculation yields 1.2, recalculate immediately.

• **Alert:** Pearson's r measures only linear relationships; if two variables have a perfect curvilinear relationship (e.g. $Y = X^2$), Pearson's r can equal zero.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Correlation. State the mathematical bounds between which Karl Pearson's correlation coefficient lies. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A statistical tool that measures the degree and direction of relationship between two or more variables. **[1 Mark]**

• **Bounds:** Bounded strictly between -1 and +1 (-1 <= r <= +1). **[1 Mark]**

2. 'Correlation does not imply causation.' Explain this statement with an illustrative example. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Meaning:** High correlation proves that two variables move together, but does not prove that one variable causes the change in the other. **[1 Mark]**

• **Example:** High correlation between ice cream sales and drowning deaths in summer is caused by hot weather (a third common factor), not by ice cream causing drowning. **[1 Mark]**

3. Describe the visual patterns in a Scatter Diagram for: (a) Perfect Positive Correlation, (b) Perfect Negative Correlation, (c) Zero Correlation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Perfect Positive Correlation:** All plotted points lie exactly on a single straight line rising upwards from bottom left to top right (r = +1). **[1 Mark]**

• **(b) Perfect Negative Correlation:** All plotted points lie exactly on a single straight line falling downwards from top left to bottom right (r = -1). **[1 Mark]**

• **(c) Zero Correlation:** Points are scattered randomly in a circular cloud showing zero discernible directional pattern (r = 0). **[1 Mark]**

4. State two merits and one limitation of Spearman's Rank Correlation Coefficient. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Merit 1:** Ideal for measuring qualitative attributes like beauty, honesty, and intelligence that can be ranked but not measured numerically. **[1 Mark]**

• **Merit 2:** Simpler and faster to compute than Pearson's product moment method. **[1 Mark]**

• **Limitation:** Not suitable for large grouped frequency distributions with tied ranks. **[1 Mark]**

5. Ranks given by two judges to 6 competitors in a dance contest are: (1, 2), (2, 1), (3, 4), (4, 3), (5, 6), (6, 5). Calculate Spearman's Rank Correlation Coefficient. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Differences in Ranks (D = R1 - R2):** -1, +1, -1, +1, -1, +1. **[1 Mark]**

• **Sum of D squared:** 1 + 1 + 1 + 1 + 1 + 1 = 6. N = 6. **[1 Mark]**

• **Formula & Computation:** r_s = 1 - (6 * Sum of D^2) divided by (N * (N^2 - 1)) = 1 - (6 × 6) divided by (6 × 35) = 1 - 36 divided by 210 = 1 - 0.171 = +0.829 (High positive correlation). **[1 Mark]**

6. Explain the property that Karl Pearson's coefficient of correlation is independent of Change of Origin and Change of Scale. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Change of Origin:** Adding or subtracting a constant from all X and Y values does not alter the correlation coefficient. **[1.5 Marks]**

• **Change of Scale:** Multiplying or dividing X and Y by any positive constant leaves r completely unchanged, proving r is a pure dimensionless number. **[1.5 Marks]**

7. Calculate Karl Pearson's coefficient of correlation between X and Y from the following paired data: X: 2, 4, 6, 8, 10; Y: 1, 3, 5, 7, 9. Interpret the result. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **Means:** Mean X = (2+4+6+8+10) divided by 5 = 6. Mean Y = (1+3+5+7+9) divided by 5 = 5. **[1 Mark]**

• **Deviations x = (X - 6):** -4, -2, 0, +2, +4 (Sum of x^2 = 16 + 4 + 0 + 4 + 16 = 40). **[1 Mark]**

• **Deviations y = (Y - 5):** -4, -2, 0, +2, +4 (Sum of y^2 = 40). Sum of x*y = 16 + 4 + 0 + 4 + 16 = 40. **[1 Mark]**

• **Calculation & Verdict:** r = 40 divided by sqrt(40 * 40) = 40 divided by 40 = +1.0. Interpretation: Perfect positive linear correlation. **[1 Mark]**